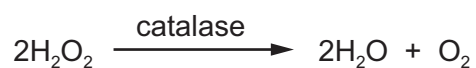


- 13** Yeast contains the enzyme catalase which catalyses the breakdown of hydrogen peroxide (H_2O_2) as shown.



Yeast was added to a solution of hydrogen peroxide and the total volume of oxygen released was recorded every 30 seconds for 2 minutes. All other variables were standardised.

The data is shown in the table.

time / s	total volume of O_2 / cm^3
30	157
60	251
90	283
120	285

What explains the pattern of the data?

- A** The rate of reaction increases as more enzyme–substrate complexes are formed.
- B** The rate of reaction increases as the enzyme reaches its maximum velocity (V_{max}).
- C** The volume of oxygen released decreases as the enzymes begin to denature.
- D** The volume of oxygen released decreases as more substrate is converted into product.